# Exercise 4

- Particle program: Simulation of particles moving randomly on a 2D grid
- Goal: Calculate the average and maximum number of particles per chare, and print them periodically
  - But do not stop the simulation or synchronize to do this, use asynchronous methods to calculate and deliver this information

# Exercise 4 details

- Grid has coordinates 100x100
- If the grid is kxm chares, each chare has a local portion of size (100/k)x(100/m)
- With n particles, if a chare is in the xth row and yth column, it generates k+(k(x+y)/(km)) particles locally
  - This creates load imbalance
- In each iteration, randomly perturb each particle in a chare
  - If the new coordinates map to another chare, you will need to send that particle to the other chare
  - If the coordinate is out of bounds, revert to the previous coordinates